Objective M6.4.2: Multiply decimals using the standard algorithm.
6.NS.B.3 Fluently add, subtract, multiply, and divide multi-digit decimals using the standard algorithm for each operation.

M6.4.2 Fellow Implementation Notes & Key

Key Points

(1) The standard algorithm for multiplying decimals is based on the concepts of place value, partial products, and properties of operations.
(2) Multiplying by 10 shifts the decimal one place right, while multiplying by 0.1 shifts the decimal one place left.
(3) The sum of the number of decimal places in the factors is equal to the number of decimal places in the product.

Lesson Overview

Students multiply decimals using properties of operations and the standard algorithm. They recognize that the number of decimal places in the product is equal to the sum of the number of decimal places in the factors, as shown at right. Students use estimation to check that answers are reasonable.

$$(4.\underbrace{03}_{1\,2}) \cdot (2.\underbrace{9}_{3})$$

$$\begin{array}{r} 295 \\ \times\quad 41 \\ \hline 295 \\ +\ 1800 \\ \hline 12.\underbrace{095}_{3\ 2\ 1} \end{array}$$

Potential Misconceptions	*Tutoring Strategies*	*Checks for Understanding*
Students may confuse the standard algorithm for adding decimals with the standard algorithm for multiplying decimals. They may attempt to line up the decimals and "bring down" the decimal point.	Use fractions, estimation, or partial products to show that multiplying decimals in this way will not result in a reasonable product.	• Do the place value of the factors need to be aligned to multiply decimals? Why not? • Is our product reasonable? How do we know?

Differentiation strategies:

(1) Provide graph paper for students to organize their work.
(2) Provide a multiplication chart for students who are not yet fluent with multiplication facts.
(3) Provide the anchor chart below for students who are not yet fluent with multiplying by powers of 10.

Power of 10	×1,000	×100	×10	×0.1 or ÷10	×0.01 or ÷100	× 0.001 or ÷1,000
Effect on decimal	→ 3 places	→ 2 places	→ 1 place	← 1 place	← 2 places	← 3 places

Key terms: factor, product, partial products

Note about additional zeros: When students write additional zeros before the first digit, some students may call it "adding a 0." Try to avoid using this phrase yourself (we aren't adding anything) and use the more accurate "annexing a zero" or "writing additional zeros."

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ELL/EML Supports	
Definitions & examples	Factor (a number being multiplied) Product (the result of multiplying two or more numbers) $0.123 \cdot 100 = 12.3$ • Point to the *factors*. • Point to the *product*.
Notes	Emphasizing the sound "-ths" when saying decimals can be especially helpful for students to distinguish decimals from whole numbers (e.g., tenths from tens). In some countries, commas and periods are sometimes used in reverse. For example, 1,2 means one and two tenths, and 1.200 means one thousand two hundred. You may need to clarify this for students. Students may have difficulty comprehending situations in word problems. Use the 3-Read Strategy to help students make sense of the problem. • What is happening in the problem? • What are the quantities in the situation and how are they related? • What is the problem asking you to find?
Synonym	Find the product (multiply)
Sentence frames/starters	The product is ___. To find the product, first I… Then I… To find the area, first I… Then I… To solve this problem, first I… Then I…

Note on opening and closings: We are working to include warm starts and optimistic closures in lessons and activities. These can increase student engagement and allows students to reflect on their learning. You are encouraged to implement these but can choose what feels best for your students.

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Opening

(See SEL Snacks: CASEL Reflection Prompts)

1. Write 2 words to describe how you are feeling today.

Image Source: Canva

Answers will vary.

(See SEL Snacks: Which One Doesn't Belong)

2. Which one doesn't belong? Be prepared to explain why.

A 0.04	B 0.4
C $\frac{4}{100}$	D $\frac{1}{10} \cdot \frac{4}{10}$

Answers will vary. Sample responses:
A. It is the only one with two digits after the decimal point.
B. It is the only one that does not have the value 4 hundredths.
C. It is the only one written as a fraction with the denominator 100.
D. It is the only one that is a multiplication expression.

Note: Click on the link above for the steps and tips to facilitate WODB.

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Optional:

Background Skills/Do Now

(See F1.15b Multiplying Decimals by Powers of 10)

1. Fill in each blank with the value that makes the statement true.

a) 12.3 · __10__ = 123

b) 0.123 · 100 = __12.3__

c) 123 · 0.01 = __1.23__

(See F1.1a Add and Subtract Whole Numbers with Regrouping)

2. What is the product of 269 · 43? Show your work.

$$\begin{array}{r} 269 \\ \times\ \ 43 \\ \hline 807 \\ +\ 10760 \\ \hline 11567 \end{array}$$

11.567

(See Lesson MSF.1.3 Convert Fractions and Decimals)

3. a) Convert each fraction to a decimal.

i. $\frac{4}{10}$

0.4

ii. $\frac{8}{100}$

0.08

b) Convert each decimal to a fraction.

i. 0.9

$\frac{9}{10}$

ii. 0.11

$\frac{11}{100}$

4. *Spiral Review or Review from Yesterday*

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Opening Critical Thinking Task

Purpose of the opening task: Recognize that the number of decimal places in the product is equal to the sum of the number of decimal places in the factors.

Evelyn applied her understanding of fractions to find the product of 0.2 and 0.3. Her work is shown below.

$$(0.2) \cdot (0.3) = \frac{2}{10} \cdot \frac{3}{10} = \frac{6}{100} = 0.06$$

1. Explain the steps Evelyn took to find the product of 0.06.

She converted the decimals to fractions, multiplied the fractions, and then converted the product to a decimal.

- Why does (0.2) · (0.3) equal $\frac{2}{10} \cdot \frac{3}{10}$?
- Why does $\frac{2}{10} \cdot \frac{3}{10}$ equal $\frac{6}{100}$?
- Why does $\frac{6}{100}$ equal 0.06?

2. Use Evelyn's method to calculate each product below.

a) (0.2) · (0.03)

$(0.2) \cdot (0.03) = \frac{2}{10} \cdot \frac{3}{100} = \frac{6}{1{,}000} = 0.006$

b) (0.02) · (0.03)

$(0.02) \cdot (0.03) = \frac{2}{100} \cdot \frac{3}{100} = \frac{6}{10{,}000} = 0.0006$

- How are the factors in these problems related? *Same digits of 2 and 3 but different place values*
- How are the products related? *Same digit of 6 but different place values*

3. Look carefully at the number of decimal places in the factors and the number of decimal places in the product in all three multiplication problems. Describe the pattern you notice.

The sum of the number of decimal places in the factors is equal to the number of decimal places in the product. For example, in part b) of question 2, there are four total decimal places in the factors and four decimal places in the product.

- In part a) of question 2, what is the sum of the number of decimal places in the factors? How many decimal places are in the product? What about in part b)?

4. Use your conclusion from the previous problem to correctly place the decimal point in each product below, given that 96 · 437 = 41,952.

a) (0.96) · (43.7) = 4 1 9 5 2

41.952

b) (9.6) · (0.437) = 4 1 9 5 2

4.1952

- How did you apply your conclusion from the previous question to answer this question?
- *Challenge:* What would be another related multiplication problem where we could find the product without doing any calculations?

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CTT to Example Connection: In the CTT, students explored how the number of decimal places in the product is equal to the sum of the number of decimal places in the factors. They will apply this rule as they multiply decimals in the examples and practice problems.

- In decimal multiplication problems, how are the number of decimal places in the product related to the number of decimal places in the factors?

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Example 1

Multiply.

$(1.2) \cdot (1.1)$

1.32

Method 1: Fractions

$1\frac{2}{10} \cdot 1\frac{1}{10} = \frac{12}{10} \cdot \frac{11}{10} = \frac{132}{100} = 1.32$

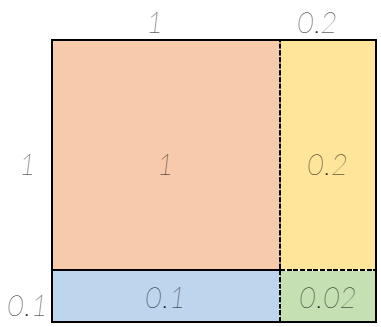
Method 2: Decimals

$12 \cdot (0.1) \cdot 11 \cdot (0.1) =$

$12 \cdot 11 \cdot (0.1) \cdot (0.1) =$

$132 \cdot (0.01) = 1.32$

Method 3: Partial products



$$\begin{array}{r} 1.2 \\ \times\ 1.1 \\ \hline 0.02 \\ 0.10 \\ 0.20 \\ +\ 1.00 \\ \hline 1.32 \end{array}$$

$(0.2) \cdot (0.1)$
$1 \cdot (0.1)$
$1 \cdot (0.2)$
$1 \cdot 1$

Purpose of Example: Multiply decimals using place value and properties of operations.

Tutoring Strategies:
Model the method(s) that students have seen in class. These methods help build conceptual understanding of the standard algorithm, which is used in Example 2.

Method 1: Fractions
Have students rewrite each factor as a fraction, convert to improper fractions, multiply, and convert the product to a decimal.

- When going from $\frac{132}{100}$ to 1.32, how does dividing by 100 affect the placement of the decimal?

Method 2: Decimals
Have students rewrite each factor as a whole number multiplied by 0.1, use the commutative property to rearrange the expression, and multiply.

- Why can we rearrange the order of the factors?
- Why does multiplying by 0.01 move the decimal two places left? *equivalent to dividing by 100*
- How would the product change if we multiplied 0.12 by 1.1 instead? *the decimal point would move one place to the left*

Method 3: Partial products
Have students create an area model and label the sides of the smaller rectangles. Students should multiply the side lengths to find the area of each smaller rectangle and add the partial products to find the product.

- How do the calculations connect to the model?
- Why do we add the partial products?
- Why do we need to align the decimal points before adding?

If students have solved the problem using more them one method, have them compare and contrast the methods.

- How are the methods similar? How are they different?
- Which method is easiest to understand? Why?

Quick check: Use at least one method to multiply (0.25) · (0.3). *0.075*

Differentiation: Students can use physical or virtual base-ten manipulatives.

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Example 2

Multiply.

a) (2.95) · (4.1)

(2.95) · (4.1)

1 2 3

$$\begin{array}{r} 295 \\ \times\ 41 \\ \hline 295 \\ +\ 11800 \\ \hline 12.095 \end{array}$$

3 2 1

12.095

b) (0.51) · (0.032)

(0.51) · (0.032)

1 2 3 4 5

$$\begin{array}{r} 51 \\ \times\ 32 \\ \hline 102 \\ +\ 1530 \\ \hline .01632 \end{array}$$

5 4 3 2 1

0.01632

Purpose of Example: Multiply decimals using the standard algorithm.

Tutoring Strategies:

Differentiation: Some students may benefit from using graph paper to organize their work. Additionally, a multiplication table may be helpful for students who are not yet fluent in multiplication facts.

a) Have students count and record the number of decimal places in the factors. Then have students multiply (295) · (41) using the standard algorithm for whole number multiplication.

- How can we turn 2.95 into a whole number? *multiply by 100 (move decimal right twice)*
    - What about 4.1?
- When we turn 2.95 into 295 and 4.1 into 41, what are we doing mathematically each time we move the decimal to the right? *multiplying by 10*
    - Why did we move the decimals in the factors? *make them whole numbers so we can multiply*
- *Challenge:* How many times greater will the product of 295 · 41 be than the product of 2.95 · (4.1)? How do we know? *1,000. We moved the decimal 3 times.*

Remind students that the sum of the decimal places in the factors is equal to the number of decimal places in the product. Move the decimal 3 places to the left in the product to place the decimal between the 2 and 0.

- Once we get a product of 12095, what are we doing mathematically each time we move the decimal to the left? *dividing by 10*
- Is our product reasonable? How do we know? *Yes, if we round each factor to the nearest whole, we get 3 · 4. 3 · 4 = 12, which is close to 12.095.*
- *Quick check:* What is the product of (29.5) · (4.1)? *120.95*

b) Have students count and record the five decimal places in the factors and then multiply 51 · 32. Count from the right of the digits in the product five times. Show students that they need to annex a zero to the left of digits in the product for the number of decimal places in the product to equal the number of decimal places in the factors.

- Why did we need to move the decimal five places left in the product?
- What is happening mathematically each time we move the decimal left?
- Is our product reasonable? Why? *Yes, the product will be smaller than either factor because both factors are less than 1.*
- Why did we need to annex a zero in part b), but not in part a)?
    - In general, when multiplying decimals, when will we need to annex one or more zeros in our product?

Potential misconception: Students may confuse the standard algorithm for adding decimals with the standard algorithm for multiplying decimals and "bring down" the decimal into the product.

- Do the place values of the factors need to be aligned to multiply decimals? Why not?

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Practice Problems

Level 1

1. a) The correct digits of each product are given below. Place the decimal in the correct place in each product.

i. (87.49) · 32 = 2 7 9 9 6 8
2799.68

ii. (13.42) · (21.75) = 2 9 1 8 8 5 0
291.885

iii. (24.85) · (19.3) = 4 7 9 6 0 5
479.605

b) Choose one of the problems above and explain how you know you placed the decimal correctly in the product.
Answers will vary. Sample response:
In ii., there are four total digits to the right of the decimal in the factors, so there must be four digits behind the decimal in the product.

2. Fill in each blank using the word bank below.

sum	product	factor
quotient	one	tenth
hundredth	two	decimal

In the multiplication problem (5.9) · (7.4) = 43.66, 5.9 is a _*factor*_ and 43.66 is the _*product*_.

Multiplying by one _*tenth*_ moves the decimal point one place to the left.

Multiplying by one hundredth moves the decimal point _*two*_ places to the left.

EML note: Not all words are used.

3. Which equation is true?

A. (219.8) · (4.3) = 9.4514

B. (21.98 · (0.43) = 945.14

C. (0.2198) · (43) = 945.14

D. (21.98) · (4.3) = 94.514

4. Which expression doesn't belong? Explain your reasoning.

A. 4 · (0.9)

B. 4 · 9 · (0.1)

C. 36 · $\frac{1}{10}$

D. 3.6

Students can argue any choice doesn't belong based on their reasoning. Sample responses:
A: The only expression with 0.9 as a factor.
B: The only expression with three factors.
C: The only expression with a fraction.
D: The only expression with one number.

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5. Find each product. Show your work.

a) (2.43) · (4.8)

2 decimal places + 1 decimal place = 3 decimal places

$$\begin{array}{r} 2\ 4\ 3 \\ \times\quad 4\ 8 \\ \hline 1\ 9\ 4\ 4 \\ +\ 9\ 7\ 2\ 0 \\ \hline 1\ 1.\ \underbrace{6\ 6\ 4}_{3\ 2\ 1} \end{array}$$

11.664

b) (0.053) · (92.7)

3 decimal places + 1 decimal place = 4 decimal places

$$\begin{array}{r} 9\ 2\ 7 \\ \times\quad 5\ 3 \\ \hline 2\ 7\ 8\ 1 \\ +\ 4\ 6\ 3\ 5\ 0 \\ \hline 4.\ \underbrace{9\ 1\ 3\ 1}_{4\ 3\ 2\ 1} \end{array}$$

4.9131

c) (5.28) · (0.31)

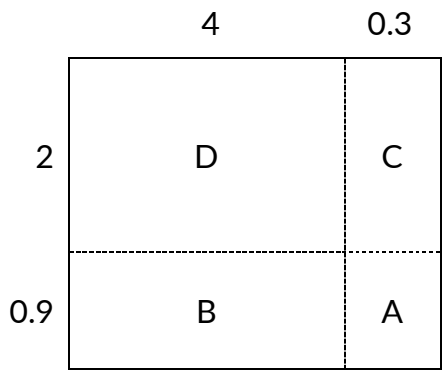
2 decimal places + 2 decimal places = 4 decimal places

$$\begin{array}{r} 5\ 2\ 8 \\ \times\quad 3\ 1 \\ \hline 5\ 2\ 8 \\ +\ 1\ 5\ 8\ 4\ 0 \\ \hline 1.\ \underbrace{6\ 3\ 6\ 8}_{4\ 3\ 2\ 1} \end{array}$$

1.6368

6. A rectangle has been partitioned into four smaller rectangles.

a) Write the letter of the smaller rectangle area that can be represented by each numerical expression.



i. 2 · (0.3)

C

ii. 2 · 4

D

iii. 4 · (0.9)

B

iv. (0.3) · (0.9)

A

b) How could we use the area of the four smaller rectangles to find the area of the big rectangle?

We could add the areas of the four smaller rectangles to find the area of the big rectangle.

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7. Use the product of 5 and 7 to find $(0.05) \cdot (0.007)$.

Since there is a total of 5 decimal places in the factors, we can find $5 \cdot 7$ and divide the product by 100,000.
$5 \cdot 7 = 35$
$35 \div 100{,}000 = 0.00035$

8. Eryn ran 3.72 miles during track practice. Kaylah ran 2.5 times the distance that Eryn ran. How many miles did Kaylah run?

2 decimal places + 1 decimal place = 3 decimal places

$$\begin{array}{r} 3\ 7\ 2 \\ \times\quad 2\ 5 \\ \hline 1\ 8\ 6\ 0 \\ +\ 7\ 4\ 4\ 0 \\ \hline 9.\ \underbrace{3\ 0\ 0}_{3\ \ 2\ \ 1} \end{array}$$

9.3 miles

9. Find each product below, given that $39 \cdot 23 = 897$.

a) $(3.9) \cdot (2.3)$
8.97

b) $0.039 \cdot 23$
0.897

c) $(0.39) \cdot (0.23)$
0.0897

10. Match each expression on the left with its product.

Column A	**Column B**
a) $(32.5) \cdot (2.41)$ *iii. 78.325*	i. 0.78325
b) $(0.325) \cdot (24.1)$ *iv. 7.8325*	ii. 783.25
c) $(32.5) \cdot (24.1)$ *ii. 783.25*	iii. 78.325
d) $(3.25) \cdot (0.241)$ *i. 0.78325*	iv. 7.8325

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11. Jayden used place value and properties of operations to find the product of (2.1) · (0.03). He accidentally left some parts of his work blank.

(2.1) · (0.03) =

21 · (0.1) · 3 · □ =

63 · □ =

0.063

a) What number must go in the first box for his work to be correct?
0.01

b) What number must go in the second box for his work to be correct?
0.001

12. A cafeteria table is 12.04 feet long and 5.9 feet wide.

a) Use estimation to determine the area of the table without calculating it exactly.

(12.04) · (5.9)

A. 7,103.6 feet squared

B. 710.36 feet squared

C. 71.036 feet squared

D. 7.1036 feet squared

b) Explain your estimation strategy.
Answers will vary. Sample response:
I rounded 12.04 to 12 and 5.9 to 6 and then multiplied 12 by 6 to get 72, which is very close to 71.036.

13. a) Andrea tried to used fraction multiplication to help her find the product of (1.1) · (0.9). Her work is shown below. In what step did she make her first error?

Step 1	$\frac{11}{100} \cdot \frac{9}{10} =$
Step 2	$\frac{99}{1{,}000} =$
Step 3	0.099

A. Step 1

B. Step 2

C. Step 3

D. There is no error.

b) Explain Andrea's error.
Andrea wrote 1.1 as $\frac{11}{100}$, but 1.1 is equal to $1\frac{1}{10}$ or $\frac{11}{10}$.

14. a) For each problem, use estimation to determine whether the product will be greater than 1 or less than 1.

i. (1.2) · (2.3)
greater than 1

ii. (0.9) · (0.85)
less than 1

iii. (1.05) · (3.09)
greater than 1

iv. (0.875) · (0.625)
less than 1

b) Explain your answer to part iv.
Answers will vary. Sample response:
The product of (0.875) · (0.625) must be less than 1 because both factors are less than 1.

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Level 2

15. Abigail tried to find the product of 2.9 and 7.3. Her work and explanation are shown below.

$$2.9 \cdot 10 = 29$$
$$7.3 \cdot 10 = 73$$
$$29 \cdot 73 = 2{,}117$$
$$2{,}117 \div 10 = 211.7$$

Because I multiplied each factor by 10 to make them into whole numbers, I divided the product of 29 and 71 by 10 to get my answer.

a) Use estimation to explain why Abigail's answer is not reasonable.

Answers will vary. Sample response:
2.9 is close to 3 and 7.1 is close to 7, so the product should be close to 21.

b) Help Abigail to understand and correct her error.

Abigail divided 2,117 by 10, but she should have divided by 100 because 2,117 is 100 times greater than the product of 2.9 and 7.3.

16. Multiply.

a) (0.46) · (83.512)

2 decimal places + 3 decimal places = 5 decimal places

$$\begin{array}{rrrrrrr} & & & 8 & 3 & 5 & 1 & 2 \\ & & \times & & & & 4 & 6 \\ \hline & & 5 & 0 & 1 & 0 & 7 & 2 \\ + & 3 & 3 & 4 & 0 & 4 & 8 & 0 \\ \hline & 3 & 8. & 4 & 1 & 5 & 5 & 2 \\ & & & \underbrace{}_{5} & \underbrace{}_{4} & \underbrace{}_{3} & \underbrace{}_{2} & \underbrace{}_{1} \end{array}$$

38.41552

b) (0.048) · (0.39)

3 decimal places + 2 decimal places = 5 decimal places

$$\begin{array}{rrrrr} & & 4 & 8 \\ & \times & 3 & 9 \\ \hline & 4 & 3 & 2 \\ + & 1 & 4 & 4 & 0 \\ \hline .0 & 1 & 8 & 7 & 2 \\ \underbrace{}_{5} & \underbrace{}_{4} & \underbrace{}_{3} & \underbrace{}_{2} & \underbrace{}_{1} \end{array}$$

0.01872

c) (12.4) · (4.35)

1 decimal place + 2 decimal places = 3 decimal places

$$\begin{array}{rrrrrr} & & & 1 & 2 & 4 \\ & & \times & 4 & 3 & 5 \\ \hline & & & 6 & 2 & 0 \\ & & 3 & 7 & 2 & 0 \\ + & 4 & 9 & 6 & 0 & 0 \\ \hline & 5 & 3. & 9 & 4 & 0 \\ & & & \underbrace{}_{3} & \underbrace{}_{2} & \underbrace{}_{1} \end{array}$$

53.94

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17. a) Which problems will have the same product as (4.73) · (8.61)? Select each correct answer.

A. (47.3) · (0.861)

B. (0.473) · (0.861)

C. (47.3) · (86.1)

D. (0.473) · (86.1)

E. (0.473) · 861

b) Explain how you selected one of your answer choices.

Answers will vary. Sample response:
Choice A: Both (4.73) · (8.61) and (47.3) · (0.861) have a total of 4 decimal places in their factors.

18. Malik tried to use partial products to find the product of 6.4 and 3.7. His work is shown below. Help Malik to understand and correct his error.

(6.4) · (3.7)

6 · 3 = 18

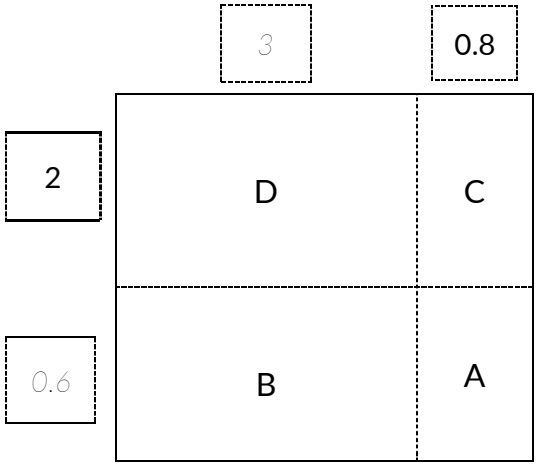
(0.4) · (0.7) = 0.28

18 + 0.28 = 18.28

Malik only found and added two of the partial products. He should have also multiplied 6 · (0.7) and 3 · (0.4) and then added all four partial products.
6 · (0.7) = 4.2
3 · (0.4) = 1.2
18 + 0.28 + 4.2 + 1.2 = 23.68

19. The area model below will be used to find the product of (3.8) · (2.6).

a) Label the missing side lengths of the area model.



b) Find the area of each of the smaller rectangles in square units.

A = 0.48 B = 1.8 C = 1.6 D = 6

c) Use the areas of the smaller rectangles to find the product of (3.8) · (2.6).

0.48 + 1.8 + 1.6 + 6 = 9.88

20. Ricardo tried to find the product of 0.04 and 1.456. His work is below.

```
  1 2 2
  1.4 5 6
× 0.0 4
  5.8 2 4
```

Identify and correct the error Ricardo made.

Ricardo aligned the decimals, as we would do in an addition problem. Instead, he should have found the product of 1456 and 4, which is 5824. Since the sum of the decimal places in the factors is 5, he can divide 5824 by 10,000 to get 0.05824.

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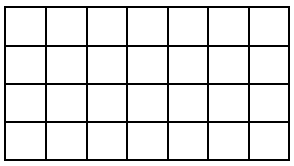
21. The area model below represents the product of 0.4 and 0.7.



a) What is the side length of each square in the model? Explain how you know.

0.1, since there are 7 units across the area model representing 0.7.

b) What is the area of each square in the model?

0.01

c) What is (0.4) · (0.7)?

0.28

22. At the local market, tomatoes are on sale for $1.25 per pound. Which of the following expressions correctly represents the cost of 0.6 pounds of tomatoes?

A. (1.25) · (0.6) = 0.75

B. (125) · (6) = 75

C. (1.25) · (0.6) = 7.5

D. (1.25) · (0.6) = 0.075

23. Which expressions have the same value as (8.2) · (0.57)? Select each correct answer.

A. 82 · (0.1) · 57 · (0.1)

B. 82 · (0.1) · 57 · (0.01)

C. 82 · (0.01) · 57 · (0.01)

D. $82 \cdot \frac{1}{10} \cdot 57 \cdot \frac{1}{10}$

E. $82 \cdot \frac{1}{10} \cdot 57 \cdot \frac{1}{100}$

24. Diamond is deciding where to place the decimal after multiplying 2.48 by 7.16. She said, "There are two decimal places in each of the factors, so there will be two decimal places in the product." Do you agree? Explain your reasoning.

No. The number of decimal places in the product is equal to the sum of the number of decimal places in the factors, so there will be 4 decimal places in the product.

25. At a grocery store, blueberries cost $1.59 per pound. Kimora bought 2.5 pounds of blueberries. How much did she pay? Round your answer to the nearest cent.

2 decimal places + 1 decimal place = 3 decimal places

$$\begin{array}{r} 1\ 5\ 9 \\ \times\quad 2\ 5 \\ \hline 7\ 9\ 5 \\ +\ 3\ 1\ 8\ 0 \\ \hline 3.\ \underbrace{9\ 7\ 5}_{3\ \ 2\ \ 1} \end{array}$$

(1.59) · (2.5) = 3.975 ≈ 3.98.

$3.98

26. Gasoline costs $2.19 per gallon. Neveah filled her car with 11.5 gallons of gas. How much did she spend? Round your answer to the nearest cent.

2 decimal places + 1 decimal place = 3 decimal places

$$\begin{array}{r} 2\ 1\ 9 \\ \times\ 1\ 1\ 5 \\ \hline 1\ 0\ 9\ 5 \\ 2\ 1\ 9\ 0 \\ +\ 2\ 1\ 9\ 0\ 0 \\ \hline 2\ 5.\ \underbrace{1\ 8\ 5}_{3\ \ 2\ \ 1} \end{array}$$

(2.19) · (11.5) = 25.185 ≈ 25.19

$25.19

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27. A rectangular field has a length of 1.25 km and a width of 0.75 km. What is the area of the field?

2 decimal places + 2 decimal place = 4 decimal places

$$\begin{array}{r} 1\ 2\ 5 \\ \times\quad 7\ 5 \\ \hline 6\ 2\ 5 \\ +\quad 8\ 7\ 5\ 0 \\ \hline 0.\ 9\ 3\ 7\ 5 \end{array}$$

4 3 2 1

0.9375 km²

28. a) How many times greater will the product of 396 · 429 be than the product of 3.96 · (4.29)?

A. 10 times greater

B. 100 times greater

C. 1,000 times greater

D. 10,000 times greater

b) Explain your reasoning for your answer in part a).

There are 4 total decimal places in the product of 3.96 · (4.29), so the product of 396 · 429 will be 10 · 10 · 10 · 10 = 10,000 times greater.

Level 3

29. Which expressions have the same value as (0.09) · (0.287)? Select each correct answer.

A. 9 · (0.01) · 287 · (0.001)

B. 9 · 287 · (0.00001)

C. 9 · (0.1) · 287 · (0.01)

D. $9 \cdot \frac{1}{10} \cdot 287 \cdot \frac{1}{100}$

E. $9 \cdot 287 \cdot \frac{1}{100{,}000}$

30. Write three multiplication expressions that have the same value as (0.03) · (0.6).

Answers will vary. Sample response:
(0.3) · (0.06)
3 · (0.006)
(0.003) · 6

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31. Find each product. Express each product as a decimal.

a) $(4.3) \cdot 2\frac{1}{2}$

4.3 · 2.5

1 decimal place + 1 decimal place = 2 decimal places

$$\begin{array}{r} 4\ 3 \\ \times\quad 2\ 5 \\ \hline 2\ 1\ 5 \\ +\quad 8\ 6\ 0 \\ \hline 1\ 0.\underbrace{7\ 5}_{2\ \ 1} \end{array}$$

10.75

b) $3\frac{4}{5} \cdot (0.62)$

3.8 · 0.62

1 decimal place + 2 decimal places =3 decimal places

$$\begin{array}{r} 3\ 8 \\ \times\quad 6\ 2 \\ \hline 7\ 6 \\ +\quad 2\ 2\ 8\ 0 \\ \hline 2.\underbrace{3\ 5\ 6}_{3\ \ 2\ \ 1} \end{array}$$

2.356

32. Zaniya multiplied two decimals to get the correct product, but she forgot to put the decimal points in her factors. Her work is shown below. What could have been the correct factors? Explain your reasoning.

$$(396) \cdot (472) = 18.6912$$

Answers will vary, but the sum of the decimal places in the factors must be 4.
Sample response: 3.96 and 4.72.

33. Use mental math and place value to find the value of n that makes each equation true.

a) $(0.4) \cdot n = 0.024$

0.06

b) $(0.07) \cdot n = 5.6$

80

c) $(0.012) \cdot n = 0.0048$

0.4

d) $25 \cdot n = 0.075$

0.003

34. a) Create your own real-world problem that could be solved by finding the product of 3.25 and 0.95.

Answers will vary. Sample response:
Ice cream costs $3.25 per pound. Sean bought 0.95 pounds of ice cream. How much did he spend?

b) Find the product of 3.25 and 0.95.

2 decimal place + 2 decimal places = 4 decimal places

$$\begin{array}{r} 3\ 2\ 5 \\ \times\quad 9\ 5 \\ \hline 1\ 6\ 2\ 5 \\ +\ 2\ 9\ 2\ 5\ 0 \\ \hline 3.\underbrace{0\ 8\ 7\ 5}_{4\ \ 3\ \ 2\ \ 1} \end{array}$$

3.0875

c) Explain what this product means in the context of your problem.

Answers will vary. Sample response:
Rounded to the nearest hundredth, Sean spent $3.09.

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35. Rico and Tyler used different methods to find the product of 1.9 and 0.46.

Rico	Tyler
$(1.9) \cdot 10 = 19$	$1.9 = \frac{19}{10}$
$(0.46) \cdot 100 = 46$	$0.46 = \frac{46}{100}$
$19 \cdot 46 = 874$	$\frac{19}{10} \cdot \frac{46}{100} = \frac{874}{1000}$
$874 \div 1{,}000 = 0.874$	$\frac{874}{1000} = 0.874$

a) Explain each student's method.

Rico multiplied the factors by 1,000 to make whole numbers, and then divided by 1,000 after finding the product of 19 and 46. Tyler converted both decimals into fractions, multiplied the fractions, and then converted the product to a decimal.

b) Use Rico's method to find the product of 2.8 and 6.7.

$(2.8) \cdot 10 = 28$
$(6.7) \cdot 10 = 67$
$28 \cdot 67 = 1{,}876$
$1{,}867 \div 100 = 18.67$

c) Use Tyler's method to find the product of 2.8 and 6.7.

$2.8 = \frac{28}{10}$

$6.7 = \frac{67}{10}$

$\frac{28}{10} \cdot \frac{67}{10} = \frac{1{,}867}{100}$

$\frac{1{,}867}{100} = 18.67$

36. Fill in each blank with always, sometimes, or never. Explain your reasoning.

a) A decimal greater than one multiplied by a decimal less than one __*sometimes*__ has a product greater than 1.

It depends on the size of the numbers. For example, $2.7 \cdot 0.9 = 2.43$, which is greater than 1, but $2.7 \cdot 0.1 = 0.27$, which is less than 1.

b) The sum of the number of decimal places in the factors is __*always*__ equal to the number of decimal places in the product.

This is true because of place value. For example, in the equation $0.2 \cdot 0.03 = 0.006$, there is a sum of three decimal places in the factors, so there are three decimal places in the product.

c) If the decimal in one factor is moved one place to the left and the decimal in the other factor is moved one place right, the product will __*never*__ change.

This is equivalent to multiplying by $\frac{10}{10}$. Multiplying by 1 does not change the value of an expression.

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37. Armando plans to cover his living room with a carpet. His living room is a rectangle that is 14.8 ft by 9.2 ft. The carpet costs $3.50 per square foot. How much will it cost to cover his living room with a carpet?

Area of rectangle = (14.8) · (9.2)
1 decimal place + 1 decimal place = 2 decimal places

```
          1 4 8
        × 9 2
        -------
          2 9 6
+ 1 3 3 2 0
-------------
  1 3 6 . 1 6
            2 1
```

136.16 · (3.50)
2 decimal places + 1 decimal place = 3 decimal places

```
        1 3 6 1 6
      × 3 5
      -----------
        6 8 0 8 0
+ 4 0 8 4 8 0
-----------------
  4 7 6 . 5 6 0
            3 2 1
```

$476.56

38. Kendall is putting a wood fence around the edges of her square garden. The side length of the garden is 8.6 feet, and wood fences cost $18.95 per linear foot. How much will she spend?

Area of square = (8.6) · 4
1 decimal place + 0 decimal places = 1 decimal place

```
      8 6
×       4
---------
  3 4 . 4
        1
```

34.4 · (18.95)
1 decimal place + 2 decimal places = 3 decimal places

```
        1 8 9 5
      × 3 4 4
      -----------
        7 5 8 0
      7 5 8 0 0
+ 5 6 8 5 0 0
-------------------
  6 5 1 . 8 8 0
            3 2 1
```

$651.88

39. Apples cost $1.75 per pound and bananas cost $0.20 each. Jamya wants to buy 3.5 pounds of apples and 6 bananas. How much will she spend? Round your answer to the nearest cent.

Cost of apples = (1.75) · (3.5)
2 decimal places + 1 decimal place = 3 decimal places

```
        1 7 5
      × 3 5
      -----------
        8 7 5
+     5 2 5 0
-----------------
      6 . 1 2 5
            3 2 1
```

Cost of bananas = (0.20) · (6)
2 decimal places + 0 decimal places = 2 decimal places

```
      2 0
×       6
---------
  1 . 2 0
        2 1
```

6.125 + 1.20 = 7.325 ≈ 7.33
$7.33

40. A recipe for one loaf of bread calls for 4.25 cups of flour. Abigail wants to make $6\frac{1}{2}$ loaves of bread. How much flour will she need? Round your answer to the nearest tenth of a cup.

$6\frac{1}{2} = 6\frac{5}{10} = 6.5$
4.25 × 6.5
2 decimal places + 1 decimal place = 3 decimal places

```
        4 2 5
      × 6 5
      -----------
      2 1 2 5
+   2 5 5 0 0
-----------------
    2 7 . 6 2 5
            3 2 1
```

27.625 ≈ 27.6
27.6 cups

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41. Marco is asked to use mental math to find 80 · (0.25). He made the statement below.

"The product is 20 because 0.25 is equivalent to $\frac{1}{4}$ and one fourth of 80 is 20."

Use Marco's strategy to help find the following products. Explain your reasoning for each.

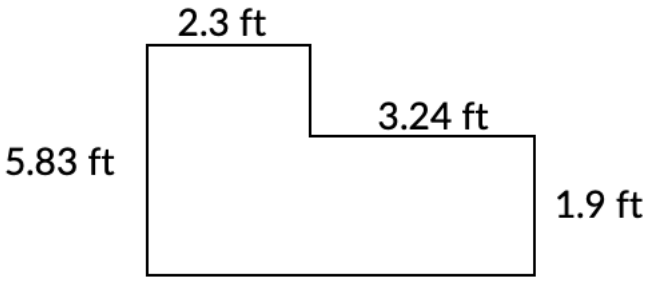
a) 80 · (2.5)
200. Explanations will vary. Sample response:
The product will be ten times greater than the product of 80 · (0.25).

b) 80 · (1.25)
100. Explanations will vary. Sample response:
80 · (0.25) = 20, so 80 · (1.25) will be 20 more than 80.

c) (0.8) · (1.25)
1. Explanations will vary. Sample response:
The problem is related to 80 · (1.25), but it will be 100 times smaller because 0.8 is 100 times smaller than 80.

Challenge

42. All of the angles in the hexagon below are right angles. What is the area of the hexagon? Show your work.



5.83 (2.3) + 3.24 (1.9) = 19.565
19.565 ft^2

43. A rectangular banner is 30 inches long and 17.5 inches wide. There are 2.54 centimeters in an inch.

What is the area of the banner, in square centimeters?

30 · 2.54 = 76.2 cm
17.5 · 2.54 = 44.45 cm
76.2 · 44.45 = 3,387.09
3,387.09 cm^2

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Closing

1. What method below do you prefer to use to multiply decimals?

Area model:

	2	0.5
1	2	0.5
0.3	0.6	0.15

$$\begin{array}{r} 2.5 \\ \times\ 1.3 \\ \hline 0^{1}.15 \\ 0.6 \\ 0.5 \\ +\ 2.0 \\ \hline 3.25 \end{array}$$

Convert to fractions:

$$2\frac{5}{10} \cdot 1\frac{3}{10} = \frac{25}{10} \cdot \frac{13}{10}$$

$$= \frac{325}{10} = 3.25$$

Decimals:

$$25 \cdot (0.1) \cdot 13 \cdot (0.1)$$

$$= 25 \cdot 13 \cdot (0.1) \cdot (0.1)$$

$$= 325 \cdot (0.01) = 3.25$$

The method I prefer is ___ because…

Answers will vary. Sample response:
The method I prefer is the area model because I like how I can break up each number to make them easier to multiply.

2. Suppose your friend is not sure where to put the decimal point in the product, shown below.

(4.13) · (2.1)

$$\begin{array}{r} 413 \\ \times\ \ 21 \\ \hline 413 \\ +\ 8260 \\ \hline 8673 \end{array}$$

What would you tell your friend to help them understand where the decimal point should go?

I would tell them…

Answers will vary. Sample response:
I would tell them to count the number of decimal places in the factors. Since 4.13 has 2 decimal places and 2.1 has 1 decimal place and 2 + 1 = 3, there should be 3 decimal places in the product. So, the product is 8.673.

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Optional:

Ticket To Leave

1. What is the product of (12.04) · (5.9)? Show your work. *(12.04) · (5.9)* *1 2 3* *2 decimal places + 1 decimal place = 3 decimal places* *1 2 0 4* *× 5 9* *1 0 8 3 6* *+ 6 0 2 0 0* *7 1 . 0 3 6* *3 2 1* *71.036*	2. a) Find each product below, given that 248 · 596 = 147,808. i. (2.48) · (5.96) *14.7808* ii. (24.8) · (59.6) *1,478.08* b) Explain how you placed the decimal in one of the products in part a). *Answers will vary. Sample response:* *In part i., the sum of the number of decimal places in the factors was 4, so I placed the decimal so there were 4 decimal places in the product.*
3. *Fellow's Choice*	4. *Fellow's Choice*

Have feedback on this lesson? Please provide it here.

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